

PENGFEI LI

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115 S 1400 E, $\diamond$ Salt Lake City, UT 84112, USA

RESEARCH INTEREST

Galaxy formation, large scale structure, and cosmology

Ly α radiation as a cosmic probe

EDUCATION

University of Utah

Sept. 2021 - Present

PhD candidate

Advisor: Prof. Zheng Zheng

Department of Physics and Astronomy

University of Science and Technology of China (USTC)

Sept. 2020 - June 2021

Graduate student

Advisor: Prof. Huiyuan Wang

Department of Astronomy, School of Physical Sciences

University of Science and Technology of China (USTC)

Sept. 2016 - June 2020

B.S. in Astronomy

Advisor: Prof. Huiyuan Wang

Wang Shouguan Talent Program in Astronomy,

Department of Astronomy, School of Physical Sciences

PUBLICATIONS AND PROJECTS

Paper in preparation

[1] **Pengfei Li** and Zheng Zheng, *Emulating extended Lyman-alpha haloes around star-forming galaxies II: modelling integral field spectroscopic observations from surface brightness profiles to radially binned spectra* (expect to be submitted in November 2026. The submission progress will be updated on Pengfei's website: <https://pengfeiliastro.github.io/publication.html>)

Paper published

[5] **Pengfei Li** and Zheng Zheng, *Analytical and fitting formulae for solutions to lyman-alpha radiative transfer equations: the effects of geometry, recoil, and velocity gradients*, 2026, MNRAS, 550, 1223

<https://doi.org/10.1093/mnras/stag1223>

[4] **Pengfei Li** and Zheng Zheng, *Emulating extended Lyman-alpha haloes around star-forming galaxies*, 2025, MNRAS, 542, 2151

<https://doi.org/10.1093/mnras/staf1354>

[3] Hui Hong, Huiyuan Wang, H. J. Mo, Ziwen Zhang, Guangwen Chen, Wentao Luo, Tinggui Wang, **Pengfei Li**, Renjie Li, Yao Yao, and Aoxiang Jiang, *Dynamical Hotness, Star Formation Quenching, and Growth of Supermassive Black Holes*, 2023, ApJ, 954, 183

<https://doi.org/10.3847/1538-4357/ace96f>

[2] Ziwen Zhang, Huiyuan Wang, Wentao Luo, H.J. Mo, Zhixiong Liang, Ran Li, Xiaohu Yang, Tinggui Wang, Hongxin Zhang, Hui Hong, Xiaoyu Wang, Enci Wang, **Pengfei Li**, and JingJing-Shi, *Host and Trigger of AGNs in the Local Universe*, 2021, A&A, 650, A155,
<https://doi.org/10.1051/0004-6361/202040150>

[1] **Pengfei Li**, Huiyuan Wang, H.J. Mo, Enci Wang, Hui Hong, *Characteristic Mass in Galaxy Quenching: Environmental versus Internal Effects*, 2020, ApJ, 902, 75
<https://doi.org/10.3847/1538-4357/abb66c>

Projects in progress

[1] Modeling Lyman-alpha surface brightness profiles from narrow band surveys, in collaboration with Dr. Satoshi Kikuta (Nara Prefectural University, Japan).

[2] Modeling FAST 21 cm observations of M31, in collaboration with Prof. Jie Wang (NAOC, China).

[3] Investigating Lyman-alpha radiative transfer in IGM, in collaboration with Prof. Shun Saito (Missouri University of Science and Technology, USA).

ACADEMIC ACTIVITIES

Oral presentation

USTC Astronomy seminar Dec. 11, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
University of Science and Technology of China, Hefei, China

Tsinghua University Galaxies & Cosmology seminar Dec. 4, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
Tsinghua University, Beijing, China

NAOC seminar Dec. 3, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
National Astronomical Observatories, Chinese Academy of Sciences, Beijing, China

Utah science talk in PHYS 4730 (Computational Physics I) Oct. 31, 2024

Talk: A Bayesian framework for modeling the gas around star-forming galaxies
University of Utah, Salt Lake City, USA

Kochel Cosmic Lyman Alpha Workshop Sept. 30-Oct. 4, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
Kochel, Munich, Germany

DHWFEST workshop

Dark, Hot, Warm, and Fuzzy matter in Space and Time July 15-17, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
University of Utah, Salt Lake City, USA

AIP CGM team meeting May 22, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
Leibniz Institute for Astrophysics Potsdam, Potsdam, Germany

MPA cosmology seminar May 14, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies

Max Plank Institute for Astrophysics, Munich, Germany

MIAPbP workshop “Some Like It Hot!”

A Journey from the Hot IGrM to the Multiphase CGM

Apr. 22-May 3, 2024

Talk: Emulating extended Lyman-alpha halos around star-forming galaxies

The Munich Institute for Astro-, Particle and BioPhysics, Munich, Germany

Workshop: Oases in the Cosmic Desert:

Understanding the Structure of the Circumgalactic Medium

Feb. 21-24, 2023

Lightening talk and poster: Understanding the Circum-galactic Medium by Emulating

Extended Lyman Alpha Halos

Arizona State University, Tempe, USA

The 22nd Guo Shoujing Galaxy Cosmology Symposium

Nov. 21-23, 2020

Talk: Characteristic Mass in Galaxy Quenching: Internal versus Environmental Effects

Zhuhai, Guangdong, China

UMass Summer Internship Talk

July 18, 2019

Quenching of Star Formation: Internal versus Environmental Effects

University of Massachusetts–Amherst, Amherst, USA

Others

The 23rd Guo Shoujing Galaxy Cosmology Symposium

May 14-17, 2021

Hangzhou, Zhejiang, China

TEACHING

Teaching assistant

Intermediate Mechanics, PHYS 3010

Fall 2026

General Physics I, PHYS 2010

Summer 2026

Physics S&E I, PHYS 2210

Spring 2026

Intermediate Mechanics, PHYS 3010

Fall 2025

General Physics I/II, PHYS 2010/2020

Summer 2025

Computational Lab, PHYS 2235

Spring 2025

Computational Physics I, PHYS 4730

Fall 2024

Tutor

Electrodynamics I/Quantum Theory I, PHYS 7110/7220 (voluntary)

Fall 2023

Electrodynamics I/Quantum Theory I, PHYS 7110/7220

Summer 2023

Electrodynamics I/Quantum Theory I, PHYS 7110/7220 (voluntary)

Fall 2022

Mentor for undergraduate research

Nathan Michael Mansfield (Undergraduate at University of Utah, now PhD student at Syracuse University).

SERVICE AND OUTREACH

Reviewer in training, Monthly Notices of the Royal Astronomical Society

Local organizing committee for DHWFEST workshop July 15-17, 2024

AstroCoffee discussion organizer, University of Utah Fall 2023 - Spring 2026

Officer of Graduate Student Advisory Committee Fall 2023 - Spring 2025

Department of Physics and Astronomy, University of Utah

REFUGES Program, Astronomy Section June-Aug., 2023

Six talks of 1.5 hours each.

Curriculum enrichment for middle and high school students from under-served community.

University of Utah, Salt Lake City, USA

Astronomy Summer Camp for High School Students June 29, 2023

Talk: New Views of the Cosmos from James Webb Space Telescope (JWST)

University of Utah, Salt Lake City, USA

Astronomy on Tap Salt Lake City Sept. 10, 2024

Talk: Cosmic Construction: from dark matter to bursting galaxies

HK Brewing, Salt Lake City, USA

HONORS AND AWARDS

Outstanding Graduate Teaching Assistant for Exceptional Fulfillment of Teaching Duties Spring, 2026

Department of Physics and Astronomy, University of Utah

Outstanding Graduate Teaching Assistant for Exceptional Fulfillment of Teaching Duties Spring, 2024

Department of Physics and Astronomy, University of Utah

Swigart Summer Research Fellowship May-Aug. 2022

Department of Physics and Astronomy, University of Utah

Outstanding Student Scholarship, Grade B (10%) 2018-2019

School of Physical Sciences, University of Science and Technology of China

Outstanding Student Scholarship, Grade C (20%) 2016-2017

School of Physical Sciences, University of Science and Technology of China

Outstanding Freshman Scholarship, Grade C (20%) 2016

School of Physical Sciences, University of Science and Technology of China

TECHNICAL SKILLS

C/C++, Python3, L^AT_EX, machine learning algorithms, Gaussian Process, Bayesian analysis